RE501 Class 4: Samples, Limit Theorems, Sigma Levels and Covariance

Debasmit Sarkar

12/10/2026

Up to now we have assumed that we *know* the distribution of a random variable: its pmf $\Pr_X(x)$ or PDF $p_X(x)$, its mean and its variance. In the laboratory it is the other way round. We do not know the distribution; we only have a handful of measured numbers. This class connects the two. We first separate the *population* (the distribution) from the *sample* (the numbers we actually measured), and see why even a perfect experiment gives scattered numbers. Two theorems, the law of large numbers and the central limit theorem, then tell us what a sample can say about its population. We then measure how rare an event is in units of σ, the language behind the six sigma and five sigma ideas of Class 3 (Section 47). Finally we extend the PDF to several random variables at once, define the covariance, and use it to propagate uncertainties.

Chapter 10: Population, sample and statistical uncertainty

49 Population and sample

Population: the complete collection of all values that *could* be observed in a random experiment, together with how often each occurs. Mathematically, the population is the probability distribution of the random variable, $\Pr_X(x)$ or $p_X(x)$.
Sample: the values that were *actually* observed, $x_1, x_2, \ldots, x_N$. The number N is the **sample size**.

Sometimes the population is a real, finite collection of objects. Sometimes it is only conceptual: "all the results this measurement could ever give if repeated forever". In physics and engineering it is almost always the second kind.

Population	Sample
All 50 000 fuel pellets in a production batch	40 pellets taken out and measured for density
All voters in a constituency	2 000 voters interviewed in an exit poll
All counts a radioactive source could give in a 1 s window	50 one-second counts actually recorded
All neutrons that could scatter from a crystal into a detector	The neutrons the detector actually registered in a run
The coolant temperature at every instant of reactor operation	One reading per minute stored by the data logger

Random sample. We will always assume that the sample is *random*: each x_i is the result of an independent repetition of the same random experiment. In the language of Class 2 (Sections 27 and 29), before we measure, the i-th reading is a random variable X_i; the $X_1, \ldots, X_N$ are independent and all have the same distribution as X. This is abbreviated **i.i.d.** (independent and identically distributed). After we measure, X_i has taken a definite value x_i. Capital letters are random variables, small letters are the numbers we wrote in the logbook.

Parameters and statistics. A number that describes the population, such as its mean or standard deviation, is a **parameter**. It is a fixed number, usually unknown. A number computed from the sample is a **statistic**. Since a different sample gives a different value, a statistic is itself a random variable.

	Population (parameter)	Sample (statistic)
What it is	the whole distribution	N observed values
Mean	$\mu = \bar{X} = E[X]$	$\hat{\mu}_N$
Standard deviation	$\sigma = \sigma_X$	s_N
Known?	usually not	yes, computed from data
Random?	no, a fixed number	yes, changes from sample to sample

A note on notation. In Classes 2 and 3 we wrote the mean of a random variable as $\bar{X}$. That is the *population* mean. From now on we also call it μ, so $\mu = \bar{X} = E[X]$ and $\sigma = \sigma_X$. Many textbooks write the *sample* mean as $\bar{x}$; to avoid any confusion with our $\bar{X}$, this handout writes it as $\hat{\mu}_N$ (the hat means "estimate of").

50 Sample mean and sample variance

From a sample $x_1, \ldots, x_N$ we compute

$$\text{Sample mean:} \quad \hat{\mu}_N = \frac{1}{N}\sum_{i=1}^{N} x_i, \qquad \text{Sample variance:} \quad s_N^2 = \frac{1}{N-1}\sum_{i=1}^{N} (x_i - \hat{\mu}_N)^2 .$$

The sample standard deviation is $s_N = \sqrt{s_N^2}$.

Compare with the population formulas of Class 2 (Sections 32 and 33): there each value was weighted by its probability $\Pr_X(x_i)$; here each measured value gets the same weight $1/N$, because each was observed once. Class 2 (Section 32) already showed the idea: if a value has probability

$\Pr_X(x_i)$, it turns up about $N \Pr_X(x_i)$ times in N repetitions, so the sample average approaches $\sum_i x_i \Pr_X(x_i)$. The law of large numbers (Section 52) makes this precise.

Why $N-1$ and not N? The deviations are measured from $\hat{\mu}_N$, which was itself calculated from the same data and so sits in the middle of the data. Deviations from $\hat{\mu}_N$ are therefore, on average, a little smaller than deviations from the true μ. Dividing by $N-1$ instead of N corrects exactly for this, so that the average of s_N^2 over many samples equals σ^2. We state this without proof. For N larger than about 20 the difference hardly matters.

51 Statistical uncertainty: randomness built into the experiment

Why does a measurement, repeated under the same conditions, give different numbers? Engineers usually list sources such as:

- **instrument uncertainty**: finite resolution, calibration error, electronic drift;
- **methodological error**: the procedure itself, e.g. sample positioning, detector dead time, background not fully subtracted;
- **modelling error**: the formula used to turn raw readings into the final quantity is only approximate;
- **fundamental limits**: limits set by nature on how well a quantity can be known even in principle.

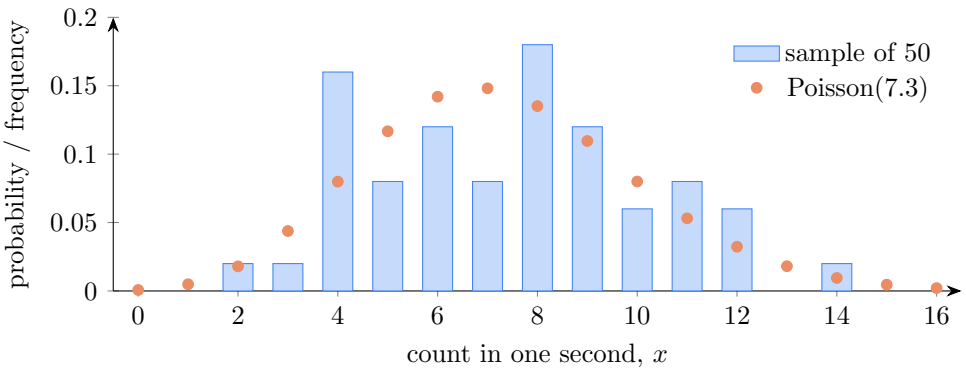
Now imagine an ideal experiment in which *all* of these are absent: a perfect detector, a perfect procedure, an exact model. In many experiments the results *still* scatter, because the phenomenon being measured is itself random. A radioactive nucleus does not decide in advance when it will decay; a neutron reaching a sample may or may not scatter into our detector. Class 3 (Section 40) showed that, for a long-lived source with many nuclei, the number of decays counted in a fixed time window is then Poisson(λ). So repeating the measurement does not reproduce the same number. It draws a new value from the population distribution.

> When the quantity measured is itself a random variable, the results of repeated measurements follow its probability distribution, even with no instrument, method, model or fundamental error at all. The standard deviation σ of that distribution is non-zero. It is called the **statistical uncertainty** of a single measurement.

Example: 50 one-second counts. A caesium-137 source placed in front of a counter gives on average $\lambda = 7.3$ counts per second. Fifty one-second counts with an ideal counter gave:

7	11	2	6	12	14	9	10	8	7
5	4	4	5	4	9	6	8	3	8
8	4	4	10	6	7	5	5	8	4
8	11	8	6	7	11	4	12	8	11
9	6	9	9	4	12	9	8	6	10

The counts range from 2 to 14, although nothing in the apparatus changed. The bars below show how often each value occurred in the sample (relative frequency = number of occurrences/50). The dots show the population, Poisson(7.3).



From the sample, $\hat{\mu}_{50} = 7.42$ and $s_{50} = 2.76$. The population values are $\mu = 7.3$ and $\sigma = \sqrt{7.3} = 2.70$ (Class 3, Section 39: Poisson mean = variance). The sample statistics are close to, but not equal to, the population parameters. Another 50 counts would give slightly different statistics.

Other experiments with built-in randomness.

- Neutron scattering: each incident neutron scatters into a given detector pixel with a small probability, so pixel counts are Poisson.
- Photon detection (shot noise): photons arrive at random times, so a photodiode current fluctuates.
- Thermal (Johnson) noise in a resistor: random motion of electrons gives a fluctuating voltage.
- Monte Carlo neutron transport codes: each simulated history is random, so every computed tally (flux, k_{eff}) comes with a statistical uncertainty.

Two properties of statistical uncertainty that distinguish it from the other sources:

1. It cannot be removed by building a better instrument. It is a property of the population, not of the apparatus.
2. It can be *reduced* by collecting more data: count for longer, or average more repetitions. How fast it shrinks is the subject of the next chapter. A calibration error, by contrast, is the same in every repetition and does not average away.

Chapter 11: Two limit laws

In this chapter $X_1, X_2, \ldots, X_N$ is a random sample (i.i.d.) from a population with mean μ and standard deviation σ, and

$$\hat{\mu}_N = \frac{X_1 + X_2 + \cdots + X_N}{N}$$

is the sample mean, now regarded as a random variable (before the measurements are made). The two laws below are stated without proof; the proofs are beyond the scope of this course.

52 The law of large numbers

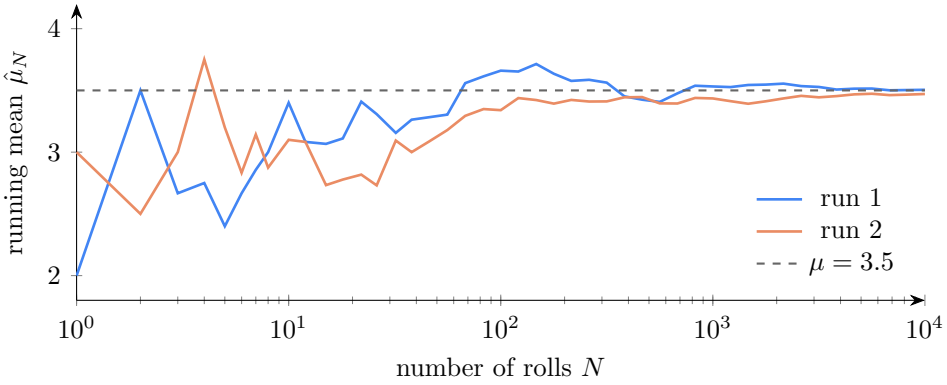
Law of large numbers. If the population mean μ exists (is finite), then as the sample size grows, the sample mean approaches the population mean:

$$\text{for every } \varepsilon > 0, \qquad \Pr(\,|\hat{\mu}_N - \mu| > \varepsilon\,) \longrightarrow 0 \quad \text{as } N \to \infty.$$

In words: *the sample mean becomes the population mean if we keep increasing the size of the sample.*

However small a tolerance ε we choose, the chance that the sample mean misses μ by more than ε can be made as small as we like by taking N large enough.

Picture. Roll a fair die again and again and keep a running average of the faces. The population mean is $\mu = (1+2+3+4+5+6)/6 = 3.5$. Two independent runs of 10 000 rolls (note the logarithmic N axis):



Early on, the running mean jumps around wildly. As N grows, both runs settle towards 3.5, along different paths.

Practical examples.

- **Empirical probability.** In Class 1 (Section 1) we estimated a probability as the fraction of trials in which an event E occurred. That fraction is the sample mean of the true/false random variable of Class 3 (Section 36), which is 1 when E occurs and 0 otherwise, and whose population mean is $\Pr(E)$. The law of large numbers is the reason the empirical fraction approaches $\Pr(E)$.
- **Averaging repeated measurements.** Averaging many readings of a quantity with statistical scatter gives a value closer and closer to the population mean.
- **Insurance and casinos.** A single policy or a single bet is unpredictable, but the average over millions is very predictable, which is why an insurer can price policies and a casino always wins in the long run.
- **Monte Carlo simulation.** A Monte Carlo code estimates a flux or a reaction rate as the average score over many simulated neutron histories. The law of large numbers guarantees that this average tends to the true value as the number of histories grows.

What the law does *not* say.

- **It does not correct the past.** After ten Tails in a row, a fair coin is *not* more likely to show Head next (the tosses are independent, Class 2 Section 27). The early excess of Tails is not cancelled; it is *diluted* by the many tosses that follow. Believing otherwise is the "gambler's fallacy".

- **It needs a finite mean.** For the Lorentzian (Cauchy) distribution,

$$p_X(x) = \frac{1}{\pi}\,\frac{\gamma}{(x-x_0)^2+\gamma^2},$$

the mean does not exist (the integral $\int x\,p_X(x)\,dx$ diverges), and the sample mean never settles down however large N is. This shape appears as the Breit–Wigner resonance in nuclear cross-sections and as the line shape in spectroscopy, so it is not merely a mathematical curiosity.

- **It does not say how fast.** For that we need the central limit theorem.

53 The central limit theorem

The law of large numbers is about $N \to \infty$. In practice N is fixed: we took 50 counts, not infinitely many. If many laboratories each took a sample of the same size N and computed its mean, the means would differ from lab to lab. How are these sample means distributed?

Central limit theorem. If the population has mean μ and finite variance σ^2, then for a fixed, large enough sample size N, the sample mean is approximately Gaussian:

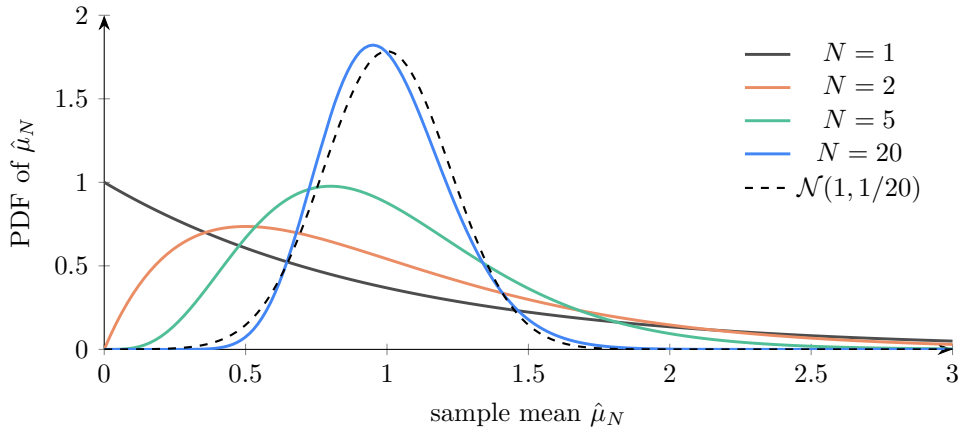
$$\hat{\mu}_N \;\approx\; \mathcal{N}\!\left(\mu,\ \frac{\sigma^2}{N}\right), \qquad \text{equivalently} \qquad X_1+\cdots+X_N \;\approx\; \mathcal{N}\big(N\mu,\ N\sigma^2\big),$$

whatever the shape of the population distribution. The approximation improves as N grows.

Three separate facts are packed in here:

1. The sample means are centred on the population mean μ.
2. Their standard deviation is $\sigma/\sqrt{N}$. This part is exact for every N; we will derive it in Section 58 once we have covariance.
3. Their distribution is a Gaussian bell, even if the population is not bell-shaped at all. This is the surprising part.

Picture. Take a strongly skewed population, the exponential distribution Exp(1) of Class 3 (Section 48), with $\mu = \sigma = 1$. Below is the exact PDF of the sample mean $\hat{\mu}_N$ for $N = 1, 2, 5, 20$. For $N = 1$ it is the population itself. As N grows, the distribution becomes narrower (width $1/\sqrt{N}$), more symmetric, and approaches the Gaussian $\mathcal{N}(1, 1/20)$ drawn dashed for $N = 20$.



Practical examples.

- **Binomial becomes Gaussian.** In Class 3 (Section 46) we saw $B(N, p)$ approach the Gaussian $\mathcal{N}\big(Np,\, Np(1-p)\big)$. The number of Heads in N tosses is the *sum* of N independent true/false variables (Class 3, Section 36), each with mean p and variance $p(1-p)$. The central limit theorem predicts exactly this Gaussian.
- **Poisson becomes Gaussian.** A count with large mean λ is the total of many independent small contributions (many nuclei, each rarely decaying), so it is nearly $\mathcal{N}(\lambda, \lambda)$ (Class 3, Section 46). This is what gives the $\pm\sqrt{N}$ error bar its 68% meaning.
- **Why measurement noise is Gaussian.** The error in a reading is often the sum of many small, independent disturbances (vibration, temperature, electronic noise, ...). Whatever their individual distributions, the sum is close to Gaussian. This is why the Gaussian is the default model for random errors.
- **Dice.** One die has a flat distribution. The sum of two dice is triangular (most likely total 7). The sum of five dice already looks like a bell.

How large is "large enough"? It depends on the population. For a symmetric population, $N \approx 5$ to 10 is often enough. For a skewed one such as the exponential, $N \approx 30$ or more may be needed. The approximation is always worst far out in the tails, which matters for the tail probabilities of Chapter 12. Like the law of large numbers, the theorem needs a finite variance: the Lorentzian fails it.

54 What the two laws mean for a measurement

The central limit theorem tells us how precisely a sample mean determines μ:

Standard error of the mean. The sample mean of N i.i.d. measurements has standard deviation

$$\sigma_{\hat{\mu}} = \frac{\sigma}{\sqrt{N}} \;\approx\; \frac{s_N}{\sqrt{N}},$$

and we report the result as $\hat{\mu}_N \pm s_N/\sqrt{N}$. Because $\hat{\mu}_N$ is nearly Gaussian, the interval $\hat{\mu}_N \pm s_N/\sqrt{N}$ contains μ with about 68% probability (Class 3, Section 46).

Example. The 50 counts of Section 51 gave $\hat{\mu}_{50} = 7.42$ and $s_{50} = 2.76$, so the mean count rate is

$$7.42 \pm \frac{2.76}{\sqrt{50}} = 7.42 \pm 0.39 \text{ counts per second.}$$

The true value 7.3 lies inside this range. Notice the difference between the two spreads: a *single* one-second count scatters by ± 2.76 (the statistical uncertainty of one measurement), but the *mean* of 50 such counts is uncertain only by ± 0.39.

The $\sqrt{N}$ law. To halve the uncertainty of a mean you need four times as much data; to reduce it tenfold, a hundred times as much. This is the same $1/\sqrt{N}$ rule as for counts in Class 3 (Section 46), and the reason Monte Carlo codes need 100 times more histories for each extra decimal place of precision.

	Law of large numbers	Central limit theorem
Question	Where does $\hat{\mu}_N$ end up as $N \to \infty$?	How is $\hat{\mu}_N$ scattered for a fixed N?
Answer	at the population mean μ	Gaussian, centred on μ, width $\sigma/\sqrt{N}$
Needs	finite mean	finite mean and variance
Gives us	confidence that averaging works	the error bar on an average

Chapter 12: How many sigmas?

55 Measuring rarity in units of σ

Thanks to the central limit theorem, many quantities in practice are close to Gaussian. For a Gaussian, the probability of landing more than k standard deviations from the mean depends only on k, not on μ or σ (Class 3, Section 46). So "how many sigmas" is a universal way of saying "how rare". We measure a deviation in units of σ:

$$z = \frac{x - \mu}{\sigma}.$$

Two kinds of tail probability are used:

- **two-sided**, $\Pr(|z| > k)$: a deviation of more than $k\sigma$ in *either* direction;
- **one-sided**, $\Pr(z > k)$: more than $k\sigma$ in *one* chosen direction. By symmetry it is half the two-sided value.

k	two-sided $\Pr(\lvert z\rvert > k)$		one-sided $\Pr(z > k)$	
1	0.317	1 in 3	0.159	1 in 6
2	0.0455	1 in 22	0.0228	1 in 44
3	2.7×10^{-3}	1 in 370	1.35×10^{-3}	1 in 741
4	6.3×10^{-5}	1 in 15 800	3.2×10^{-5}	1 in 31 600
5	5.7×10^{-7}	1 in 1.7 million	2.9×10^{-7}	1 in 3.5 million
6	2.0×10^{-9}	1 in 507 million	9.9×10^{-10}	1 in 1 billion

Each extra sigma makes the event far rarer: from 3σ to 5σ the probability drops by a factor of almost 5 000. These numbers are only as good as the Gaussian assumption, and real distributions often have fatter tails than the Gaussian (Section 53). The rows connect to Class 3 (Section 47): the two-sided 3σ and 6σ values are the scrap fractions of the rod example (0.27% and 2 per billion), and the one-sided 5σ value, 2.9×10^{-7}, is the threshold for claiming a discovery.

Chapter 13: Several random variables

56 Joint distributions

Often one random experiment produces several numbers at once: the counts in two detectors, the length and the diameter of a pellet, the slope and intercept of a fitted line. Each is a random variable on the same sample space Ω, and we need their probabilities *together*.

Discrete case. For two discrete random variables X and Y, the **joint pmf** is

$$\Pr_{X,Y}(x,y) = \Pr(X = x \text{ and } Y = y) = \Pr\left(X^{-1}(x) \cap Y^{-1}(y)\right),$$

the probability of the intersection of two events (Class 1, Section 9 for intersections; Class 2, Section 29 for preimages).

Example. Toss a fair coin three times. Let X be the number of Heads in tosses 1 and 2, and Y the number of Heads in tosses 2 and 3. Listing the 8 equally likely outcomes gives the joint pmf (entries in units of 1/8):

	$y=0$	$y=1$	$y=2$	$\Pr_X(x)$
$x=0$	1	1	0	2
$x=1$	1	2	1	4
$x=2$	0	1	1	2
$\Pr_Y(y)$	2	4	2	8

For example $\Pr_{X,Y}(1,1) = 2/8$ from the outcomes HTH and THT. Summing a row gives the distribution of X alone; summing a column gives that of Y alone. These are the **marginal** distributions; here both are $B(2, \frac{1}{2})$, as in Class 2 (Section 30).

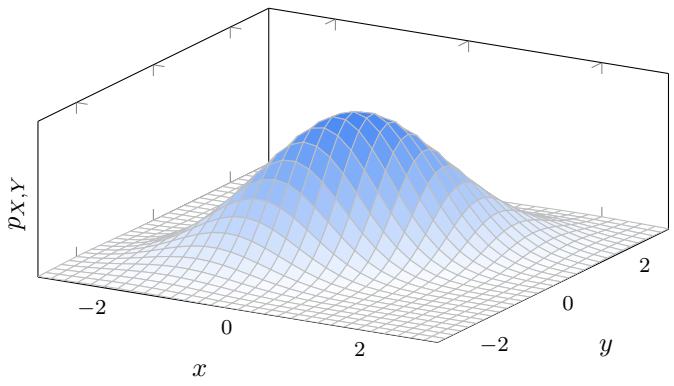
Continuous case. For two continuous random variables the **joint PDF** $p_{X,Y}(x,y)$ is a surface over the (x,y) plane. Just as $p_X(x)\,dx$ is the probability of a thin strip (Class 3, Section 43), $p_{X,Y}(x,y)\,dx\,dy$ is the probability that (X,Y) lands in a small rectangle of sides dx and dy at (x,y). Probabilities are volumes under the surface:

$$\Pr\big((X,Y) \in A\big) = \iint_A p_{X,Y}(x,y)\,dx\,dy, \qquad p_{X,Y}(x,y) \geq 0,$$

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} p_{X,Y}(x,y)\,dx\,dy = 1.$$

Marginal PDF: $\displaystyle p_X(x) = \int_{-\infty}^{\infty} p_{X,Y}(x,y)\,dy, \qquad p_Y(y) = \int_{-\infty}^{\infty} p_{X,Y}(x,y)\,dx.$

Integrating out y is the continuous version of summing a row of the table.



A two-dimensional Gaussian PDF (formula in Section 57). The hill is stretched along the diagonal: large x tends to come with large y.

Independence. In Class 2 (Section 27), events E and E' are independent when $\Pr(E \cap E') = \Pr(E)\Pr(E')$. Applied to the events $\{X = x\}$ and $\{Y = y\}$, random variables are **independent** when the joint distribution factorises:

$$\Pr_{X,Y}(x,y) = \Pr_X(x)\,\Pr_Y(y) \quad \text{(discrete)}, \qquad p_{X,Y}(x,y) = p_X(x)\,p_Y(y) \quad \text{(continuous)},$$

for all x, y. In the coin example they are not independent: $\Pr_{X,Y}(0,2) = 0$ but $\Pr_X(0)\Pr_Y(2) = \frac{2}{8}\cdot\frac{2}{8} \neq 0$. Both variables share toss 2.

More than two variables. Everything extends to n random variables $X_1, \ldots, X_n$, collected into a vector $\mathbf{X} = (X_1, \ldots, X_n)$. Its joint PDF $p_{\mathbf{X}}(x_1, \ldots, x_n)$ gives probabilities as n-fold integrals. A random sample of Chapter 11 is an example: n i.i.d. variables, whose joint PDF is the product $p_X(x_1)\,p_X(x_2)\cdots p_X(x_n)$.

57 Covariance and correlation

The mean and variance summarise one random variable. For two, we need one more number: do they tend to move together?

Covariance of X and Y, with means μ_X, μ_Y:

$$\mathrm{Cov}(X,Y) = \sigma_{XY} = E\big[(X-\mu_X)(Y-\mu_Y)\big] = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} (x-\mu_X)(y-\mu_Y)\,p_{X,Y}(x,y)\,dx\,dy$$

(for discrete variables, $\sum_x \sum_y (x-\mu_X)(y-\mu_Y)\Pr_{X,Y}(x,y)$). Expanding the product, exactly as for the variance in Class 2 (Section 34.2),

$$\mathrm{Cov}(X,Y) = E[XY] - \mu_X\,\mu_Y.$$

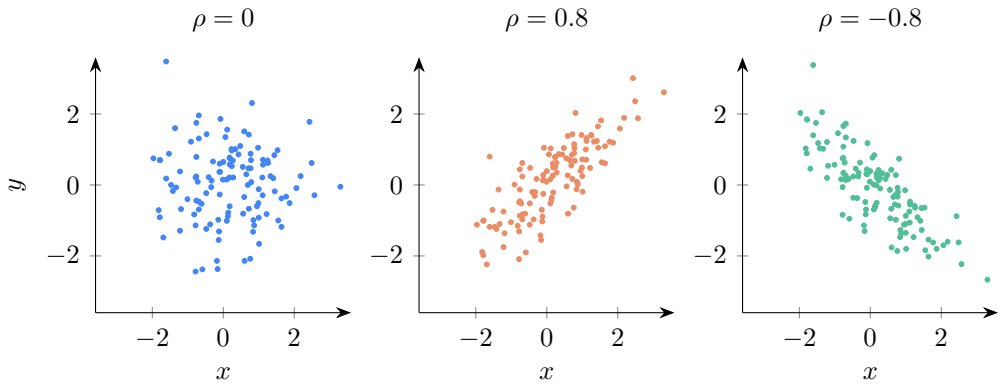
Correlation coefficient: $$\rho_{XY} = \frac{\mathrm{Cov}(X,Y)}{\sigma_X\,\sigma_Y}, \qquad -1 \le \rho_{XY} \le 1.$$

Meaning of the sign. The product $(x-\mu_X)(y-\mu_Y)$ is positive when x and y are on the same side of their means, negative when on opposite sides. So:

- $\mathrm{Cov} > 0$: when X is above its mean, Y tends to be above its mean too;
- $\mathrm{Cov} < 0$: when X is high, Y tends to be low;
- $\mathrm{Cov} = 0$: no *linear* tendency either way.

The covariance carries the units of X times the units of Y. The correlation coefficient ρ is dimensionless: $\rho = \pm 1$ means Y is an exact straight-line function of X. Note also that $\mathrm{Cov}(X,X) = \sigma_X^2$: the variance is the covariance of a variable with itself.

Each scatter plot below shows 120 draws of (X,Y), both with mean 0 and standard deviation 1:



Example: the three coin tosses. From the table in Section 56, $\mu_X = \mu_Y = 1$ and

$$E[XY] = \sum_x \sum_y x\,y\,\Pr_{X,Y}(x,y) = 1\cdot 1\cdot \tfrac{2}{8} + 1\cdot 2\cdot \tfrac{1}{8} + 2\cdot 1\cdot \tfrac{1}{8} + 2\cdot 2\cdot \tfrac{1}{8} = \tfrac{10}{8},$$

so $\mathrm{Cov}(X,Y) = \frac{10}{8} - 1\cdot 1 = \frac{1}{4}$. With $\sigma_X^2 = \sigma_Y^2 = Np(1-p) = \frac{1}{2}$ (Class 3, Section 37), $\rho_{XY} = \frac{1/4}{1/2} = 0.5$. The positive correlation comes from the shared toss 2: if it is a Head, both X and Y go up. The same thing happens in the laboratory when two measured quantities share a common ingredient, for example two spectra normalised by the same monitor count.

Independent $\Rightarrow$ uncorrelated. If X and Y are independent, $p_{X,Y} = p_X\,p_Y$ and the double integral splits:

$$\mathrm{Cov}(X,Y) = \int (x-\mu_X)\,p_X(x)\,dx \;\cdot\; \int (y-\mu_Y)\,p_Y(y)\,dy = 0\cdot 0 = 0,$$

since the first central moment is always zero (Class 2, Section 34.1).

Uncorrelated does *not* imply independent. Let X take the values $-1, 0, 1$ with probability $\frac{1}{3}$ each, and $Y = X^2$. Then Y is completely determined by X, yet $E[X] = 0$ and $E[XY] = E[X^3] = 0$, so $\mathrm{Cov}(X,Y) = 0$. Covariance detects only straight-line tendencies.

The covariance matrix. For n random variables $X_1, \ldots, X_n$ we collect all the covariances $V_{ij} = \mathrm{Cov}(X_i, X_j)$ into an $n \times n$ matrix:

$$V = \begin{pmatrix} \sigma_1^2 & \mathrm{Cov}(X_1,X_2) & \cdots & \mathrm{Cov}(X_1,X_n) \\ \mathrm{Cov}(X_2,X_1) & \sigma_2^2 & \cdots & \mathrm{Cov}(X_2,X_n) \\ \vdots & \vdots & \ddots & \vdots \\ \mathrm{Cov}(X_n,X_1) & \mathrm{Cov}(X_n,X_2) & \cdots & \sigma_n^2 \end{pmatrix}.$$

The diagonal holds the variances. The matrix is symmetric, $V_{ij} = V_{ji}$. For independent variables it is diagonal.

Every least-squares fitting program returns such a matrix for the fitted parameters; its diagonal gives their squared uncertainties, and its off-diagonal entries tell how the parameters are correlated.

For reference: the multivariate Gaussian. The Gaussian of Class 3 (Section 46) extends to two variables with means μ_X, μ_Y, standard deviations σ_X, σ_Y and correlation ρ:

$$p_{X,Y}(x,y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \times \exp\left\{ -\frac{1}{2(1-\rho^2)} \left[\frac{(x-\mu_X)^2}{\sigma_X^2} - \frac{2\rho(x-\mu_X)(y-\mu_Y)}{\sigma_X\sigma_Y} + \frac{(y-\mu_Y)^2}{\sigma_Y^2} \right] \right\},$$

and to n variables with mean vector $\boldsymbol{\mu}$ and covariance matrix V:

$$p_{\mathbf{X}}(\mathbf{x}) = \frac{1}{(2\pi)^{n/2}\sqrt{\det V}} \exp\Big(-\tfrac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^{\mathsf{T}} V^{-1} (\mathbf{x}-\boldsymbol{\mu})\Big).$$

The surface plot in Section 56 is the first formula with $\sigma_X = \sigma_Y = 1$ and $\rho = 0.6$. A Gaussian is completely fixed by its means and its covariance matrix.

58 Variance of a sum

For any random variables X, Y and constants a, b:

$$\operatorname{Var}(aX + bY) = a^2\sigma_X^2 + b^2\sigma_Y^2 + 2ab\,\operatorname{Cov}(X,Y).$$

If X and Y are independent (or merely uncorrelated), the last term vanishes and *variances add.*

Derivation. The mean of $aX + bY$ is $a\mu_X + b\mu_Y$, so

$$\begin{aligned}\operatorname{Var}(aX+bY) &= E\Big[\big(a(X-\mu_X) + b(Y-\mu_Y)\big)^2\Big] \\ &= a^2 E\big[(X-\mu_X)^2\big] + b^2 E\big[(Y-\mu_Y)^2\big] + 2ab\,E\big[(X-\mu_X)(Y-\mu_Y)\big] \\ &= a^2\sigma_X^2 + b^2\sigma_Y^2 + 2ab\,\operatorname{Cov}(X,Y).\end{aligned}$$

The standard error of the mean, at last. For an i.i.d. sample, all covariances between different X_i are zero, so the variance of the sum is the sum of the variances:

$$\operatorname{Var}\left(\frac{X_1 + \cdots + X_N}{N}\right) = \frac{1}{N^2}\big(\sigma^2 + \sigma^2 + \cdots + \sigma^2\big) = \frac{N\sigma^2}{N^2} = \frac{\sigma^2}{N},$$

which is the $\sigma/\sqrt{N}$ promised in Section 53. Independence was essential: correlated readings (for example, all affected by the same slow temperature drift) average down more slowly.

Check with the coins. $X+Y$ in the three-toss example has variance $\frac{1}{2}+\frac{1}{2}+2\cdot\frac{1}{4} = \frac{3}{2}$. Directly: $X + Y = T_1 + 2T_2 + T_3$, where T_i is 1 for a Head on toss i (variance $\frac{1}{4}$ each, independent), so $\operatorname{Var} = \frac{1}{4} + 4\cdot\frac{1}{4} + \frac{1}{4} = \frac{3}{2}$. ✓

59 Propagation of uncertainty

We rarely report a raw measurement. We compute something from it: a count rate from a count and a time, a momentum transfer from an angle and a wavelength. If the inputs are uncertain, how uncertain is the result?

Let $f(X,Y)$ be computed from two uncertain inputs. If the uncertainties are small, f is nearly linear over the range where X and Y scatter, so we may use the first-order Taylor expansion about the means:

$$f(X,Y) \approx f(\mu_X,\mu_Y) + \frac{\partial f}{\partial x}(X-\mu_X) + \frac{\partial f}{\partial y}(Y-\mu_Y),$$

with the derivatives evaluated at (μ_X, μ_Y). This is of the form constant $+\,aX + bY$, so Section 58 gives at once:

Propagation of uncertainty (first order):

$$\sigma_f^2 = \left(\frac{\partial f}{\partial x}\right)^2 \sigma_X^2 + \left(\frac{\partial f}{\partial y}\right)^2 \sigma_Y^2 + 2\,\frac{\partial f}{\partial x}\,\frac{\partial f}{\partial y}\,\mathrm{Cov}(X,Y).$$

For n inputs: $\sigma_f^2 = \sum_{i=1}^{n}\sum_{j=1}^{n} \frac{\partial f}{\partial x_i}\,\frac{\partial f}{\partial x_j}\,V_{ij}$.

This is the "law of propagation of uncertainty" of the international *Guide to the Expression of Uncertainty in Measurement* (GUM). The familiar "add in quadrature" rule is the special case of independent inputs, where the covariance term is dropped.

Example 1: background subtraction (independent inputs). A peak window holds $G = 1600$ gross counts; a background run gives $B = 900$ counts. The net signal is $S = G - B = 700$. The two counts are independent Poisson variables, so $\sigma_G^2 = 1600$, $\sigma_B^2 = 900$, and with $\partial S/\partial G = 1$, $\partial S/\partial B = -1$:

$$\sigma_S^2 = 1600 + 900 = 2500, \qquad S = 700 \pm 50.$$

The uncertainties *add* even though the counts are *subtracted*. Each count alone is uncertain by about 2.5–3%, but the net signal by 7%.

Example 2: products and ratios (independent inputs). For $f = XY$ or $f = X/Y$, dividing the formula by f^2 gives

$$\left(\frac{\sigma_f}{f}\right)^2 = \left(\frac{\sigma_X}{X}\right)^2 + \left(\frac{\sigma_Y}{Y}\right)^2 :$$

relative uncertainties add in quadrature. For a count rate $R = N/t$ with a precise timer, $\sigma_R/R = \sigma_N/N = 1/\sqrt{N}$.

Example 3: a neutron diffraction peak. The momentum transfer is $q = 4\pi \sin\theta/\lambda$. With $\lambda = 4.00 \pm 0.04$ Å and $\theta = 30.0^\circ \pm 0.1^\circ$ (independent),

$$q = \frac{4\pi \times 0.5}{4.00} = 1.571 \text{ Å}^{-1}, \qquad \frac{\partial q}{\partial \lambda} = -\frac{q}{\lambda}, \qquad \frac{\partial q}{\partial \theta} = q\cot\theta,$$

so (with σ_θ in radians, $0.1^\circ = 1.745 \times 10^{-3}$)

$$\frac{\sigma_q}{q} = \sqrt{\left(\frac{0.04}{4.00}\right)^2 + (\cot 30^\circ \times 1.745 \times 10^{-3})^2} = \sqrt{(0.0100)^2 + (0.0030)^2} = 0.0104,$$

and $q = 1.571 \pm 0.016$ Å^{-1}. The wavelength spread dominates; improving the angle measurement would hardly help.

Example 4: why the covariance term matters. A straight-line calibration $y = a + bx$ is fitted to data. The fit returns $a = 2.0 \pm 0.3$, $b = 0.50 \pm 0.05$ and correlation $\rho_{ab} = -0.9$, so $\mathrm{Cov}(a,b) = -0.9 \times 0.3 \times 0.05 = -0.0135$. (Intercept and slope of a fit are typically strongly anti-correlated: if the line is tilted a little steeper, it must start a little lower to pass through the data.) The calibrated value at $x_0 = 10$ is $y_0 = a + 10\,b = 7.0$, with $\partial y_0/\partial a = 1$ and $\partial y_0/\partial b = 10$:

$$\sigma_{y_0}^2 = 0.3^2 + 10^2 \times 0.05^2 + 2 \times 1 \times 10 \times (-0.0135) = 0.09 + 0.25 - 0.27 = 0.07,$$

so $y_0 = 7.0 \pm 0.26$. Ignoring the covariance would give $\sqrt{0.34} = 0.58$, more than twice too large. With a positive correlation, ignoring it would instead *underestimate* the uncertainty, which is the more dangerous mistake.

When is first order enough? The linear approximation needs f to be nearly straight over a range of a few σ around the means. For strongly non-linear functions or large uncertainties, one instead simulates: draw many random inputs from their joint distribution, compute f for each, and look at the spread of the results (a Monte Carlo propagation, again justified by the law of large numbers).

60 The sandwich rule

Often we compute *several* results from the same inputs, and then we also need the covariances *between the results*. Matrix notation handles all of this in one line.

Let the inputs be the vector $\mathbf{x} = (x_1, \ldots, x_n)$ with covariance matrix V_x $(n \times n)$, and the results be $\mathbf{y} = (y_1, \ldots, y_m)$ with $y_k = f_k(\mathbf{x})$. Collect the derivatives into the $m \times n$ **Jacobian matrix** J, with $J_{ki} = \partial f_k / \partial x_i$ (row k is the gradient of f_k). Then:

Sandwich rule:

$$V_y = J\, V_x\, J^{\mathsf{T}}.$$

V_y is the $m \times m$ covariance matrix of the results: its diagonal holds the variances of the results, its off-diagonal entries their covariances.

The name comes from the shape: the input covariance V_x is the filling, sandwiched between two slices of Jacobian, J and its transpose J^{T}.

It contains everything we did before. With one result ($m = 1$) and two inputs, $J = \begin{pmatrix} \partial f/\partial x & \partial f/\partial y \end{pmatrix}$ is a single row, and

$$J V_x J^{\mathsf{T}} = \begin{pmatrix} f_x & f_y \end{pmatrix} \begin{pmatrix} \sigma_X^2 & \mathrm{Cov}(X,Y) \\ \mathrm{Cov}(X,Y) & \sigma_Y^2 \end{pmatrix} \begin{pmatrix} f_x \\ f_y \end{pmatrix} = f_x^2 \sigma_X^2 + 2 f_x f_y \,\mathrm{Cov}(X,Y) + f_y^2 \sigma_Y^2,$$

writing $f_x = \partial f/\partial x$, $f_y = \partial f/\partial y$. This is exactly the propagation formula of Section 59. For Example 4 there, $J = (1 \;\; 10)$ and

$$\sigma_{y_0}^2 = \begin{pmatrix} 1 & 10 \end{pmatrix} \begin{pmatrix} 0.09 & -0.0135 \\ -0.0135 & 0.0025 \end{pmatrix} \begin{pmatrix} 1 \\ 10 \end{pmatrix} = 0.07.$$

Example: independent inputs can give correlated results. Two detectors record independent Poisson counts $A = 900$ and $B = 400$, so $V_x = \begin{pmatrix} 900 & 0 \\ 0 & 400 \end{pmatrix}$. We report the total $S = A + B$ and the difference $D = A - B$. The Jacobian is $J = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$, and

$$V_y = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 900 & 0 \\ 0 & 400 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \begin{pmatrix} 1300 & 500 \\ 500 & 1300 \end{pmatrix}.$$

So $S = 1300 \pm 36$ and $D = 500 \pm 36$, but S and D are *correlated*, with $\rho_{SD} = 500/1300 \approx 0.38$: an upward fluctuation of A raises both. Anyone who later combines S and D must use this covariance, not just the two error bars. This is why careful experiments publish the full covariance matrix of their results, not only the uncertainties.

Summary

Idea	Key formula or fact
Sample mean and variance	$\hat{\mu}_N = \frac{1}{N}\sum x_i, \quad s_N^2 = \frac{1}{N-1}\sum (x_i - \hat{\mu}_N)^2$
Statistical uncertainty	σ of the population; non-zero even for a perfect experiment if the phenomenon is random
Law of large numbers	$\hat{\mu}_N \to \mu$ as $N \to \infty$
Central limit theorem	$\hat{\mu}_N \approx \mathcal{N}(\mu, \sigma^2/N)$ for fixed large N, any population with finite σ
Standard error of the mean	$\sigma/\sqrt{N} \approx s_N/\sqrt{N}$
5σ, 6σ (one-sided)	2.9×10^{-7} (1 in 3.5 million), 9.9×10^{-10} (1 in 1 billion)
Joint PDF	$\Pr((X,Y) \in A) = \iint_A p_{X,Y}\, dx\, dy$; marginal $p_X = \int p_{X,Y}\, dy$
Covariance	$\mathrm{Cov}(X,Y) = E[(X-\mu_X)(Y-\mu_Y)] = E[XY] - \mu_X\mu_Y$; $\rho = \mathrm{Cov}\,/(\sigma_X\sigma_Y)$
Variance of a sum	$\mathrm{Var}(aX+bY) = a^2\sigma_X^2 + b^2\sigma_Y^2 + 2ab\,\mathrm{Cov}(X,Y)$
Uncertainty propagation	$\sigma_f^2 = \sum_{i,j} \frac{\partial f}{\partial x_i}\frac{\partial f}{\partial x_j} V_{ij}$
Sandwich rule	$V_y = J V_x J^{\mathsf{T}}$